

CSA1217 ARTIFICIAL INTELLIGENCE

CO1- ASSESSMENT TOOL 1

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Question 1: Water Jug Problem

Problem Statement:

You are given a 4-gallon jug and a 3-gallon jug without any measuring marks. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug by filling, emptying, and pouring water between the jugs.

Initial State

(0,0) – Both jugs are empty.

Goal State

(2,x) – The 4-gallon jug contains exactly 2 gallons.

Solution Steps

Step	Action	State (4G,3G)
1	Fill the 3-gallon jug.	(0,3)
2	Pour it into the 4-gallon jug.	(3,0)
3	Fill the 3-gallon jug again.	(3,3)
4	Pour into the 4-gallon jug until it becomes full.	(4,2)
5	Empty the 4-gallon jug.	(0,2)
6	Transfer the remaining 2 gallons into the 4-gallon jug.	(2,0)

Result

The required 2 gallons are successfully obtained in the 4-gallon jug.

Question 2: Mars Rover Agent

i. Percepts

- Terrain images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Camera images
- Obstacle detection

ii. Operating Environment

- Partially Observable
- Sequential
- Dynamic
- Continuous
- Single Agent
- Unknown Environment

iii. Actions

- Move forward
- Move backward
- Turn left/right
- Capture images
- Collect soil samples
- Analyze samples
- Transmit data to Earth
- Recharge using solar panels

iv. Performance Measure

- Safe navigation
- Maximum scientific data collection
- Accurate analysis
- Efficient energy usage
- Mission completion
- Reliable communication

v. Suitable Agent Architecture

- Utility-Based Agent
- Reason: It chooses actions that maximize mission success while balancing safety, energy consumption, and scientific objectives.

Question 3: 8-Queens Problem

The objective is to place eight queens on an 8×8 chessboard such that no two queens attack each other.

State Space

Each state represents one arrangement of queens on the chessboard.

Initial State

An empty chessboard.

Goal State

Eight queens placed without sharing the same row, column, or diagonal.

Operators

Place a queen, move a queen, or remove a queen.

Search Strategy

Backtracking Search is commonly used to explore valid configurations and backtrack whenever a conflict occurs.

Example Solution

Column → Row : (1→1), (2→5), (3→8), (4→6), (5→3), (6→7), (7→2), (8→4).

Question 4: OLA Cab Booking

Problem Solving Agent

A Goal-Based Agent is suitable because the customer's objective is to reach the destination by selecting the preferred cab type.

Pseudocode

START

Input pickup location

Input destination

Display available cab types

Customer selects a cab

IF cab is available THEN

```
    Calculate fare
    Confirm booking
ELSE
    Display "Cab Not Available"
ENDIF
Track driver location
Generate bill
Complete payment
END
```

Question 5: Uniform Cost Search

The weighted graph contains the following edges:

$S \rightarrow A=1$, $S \rightarrow G=12$, $A \rightarrow B=3$, $A \rightarrow C=1$, $B \rightarrow D=3$, $C \rightarrow D=1$, $C \rightarrow G=2$, $D \rightarrow G=3$.

Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$1 + 1 + 2 = 4$$

Conclusion

Uniform Cost Search always expands the node with the lowest cumulative path cost and therefore finds the optimal path.

Overall Conclusion

This assessment covers important Artificial Intelligence concepts including state-space search, intelligent agents, backtracking, goal-based reasoning, utility-based reasoning, and Uniform Cost Search. These techniques enable intelligent systems to solve complex real-world problems efficiently and optimally.